INCH-POUND

MIL-PRF-49471/5(ER)
2 June 1995

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X847A/U

This specification sheet is approved for use within Army Research Laboratory , Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-49471(ER).

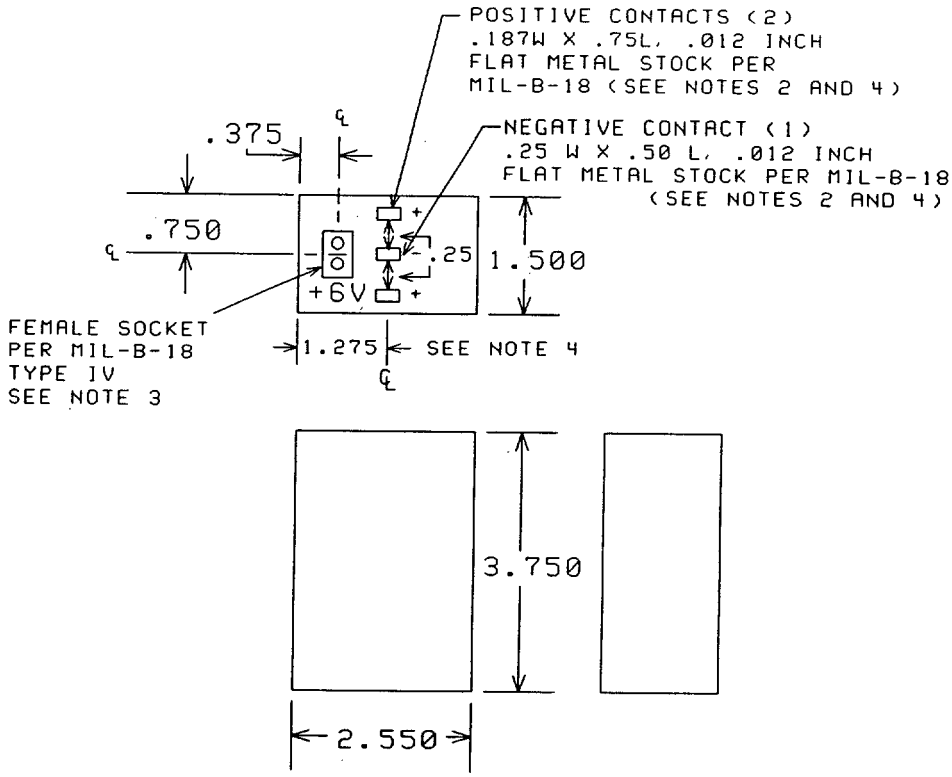


Figure 1. Battery dimensions.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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NOTES:

1. All dimensions are in inches. The tolerance shall be ±0.031 inch unless otherwise specified.

2. The dual connectors shall be wired in a parallel 6 volt circuit.

3. Type IV socket:

   a. Pin circle center shall be on the battery center line.

   b. Minimum depth of the socket well shall be 5/8 inch.

   c. Shall float 1/32 inch in all directions from the pin circle center.

4. Flat contacts:

   a. The center negative contact shall be on both battery center lines.

   b. Shall be supported and mounted so that the top surface shall not protrude above the outside surface of the battery jacket.

   c. Shall not be more than 1/8 inch below the outside surface of the battery jacket initially, during, and after contact is made with mating contact springs.

   d. Shall not have a lateral displacement of more than 1/32 inch during mating with contact springs.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 6.10 volts.

Minimum voltage for capacity: 4.00 volts.

Safety features: Applicable.

Charge protection: Applicable.

Dual Terminals: Two (2) hole female socket per MIL-B-18, Type IV; three flat contacts as per Figure 1.

Weight (maximum): 0.60 pound (272 grams).

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Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Minimum Capacity Requirements in Hours to 4.00 Volts
I	2.5
L	1.3
H	5.0
HT	4.8
LT	1.2
IT	2.3

Initial voltage delay: During the I, H, HT, and IT discharge tests initial voltages below 4.00 volts cannot exceed a 5.0 second duration. During the L and LT discharge tests initial voltage below 3.80 volts cannot exceed a 60.0 second duration.

High temperature device: Applicable.

Complete discharge device: Applicable.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for all capacity tests are specified in the basic specification.

(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 3.0 amperes at 70±5°F to 2/3 of open-circuit voltage. Each discharged cell shall then be connected in series to one fresh cell. Each string shall be discharged at 3.0 amperes at 70±5°F to 2/3 open-circuit voltage of the one fresh cell.

Batteries: For all capacity tests, except the H and HT, the batteries shall be discharged at three amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through three amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts.

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Closed circuit voltage test: Load resistance of 3 ohms shall be used. Voltage shall be above 4.00 volts within 5 seconds.

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